$$r = \frac{8\times 24 - 0\times 0}{\sqrt{8\times 36 - (0)^2}\ \sqrt{8\times 44 - (0)^2}}$$

$$r = \frac{8\times 24}{\sqrt{8\times 36}\ \sqrt{8\times 44}}$$

$$= 0.603$$

Note: The correlation coefficient computed by using direct method and short-cut method is the same.

Example 4.2

The following are the marks scored by 7 students in two tests in a subject. Calculate coefficient of correlation from the following data and interpret.

Marks in test-1	12	9	8	10	11	13	7
Marks in test-2	14	8	6	9	11	12	3

Solution:

Let x denote marks in test-1 and y denote marks in test-2.

	x_i	y_i	x_i^2	y_i^2	x_iy_i
	12	14	144	196	168
	9	8	81	64	72
	8	6	64	36	48
	10	9	100	81	90
	11	11	121	121	121
	1	12	169	144	156
	7	3	49	9	21
Total	70	63	728	651	676

$$r = \frac{n\sum_{i=1}^{n} x_iy_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{\sqrt{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}\ \sqrt{n\sum_{i=1}^{n} y_i^2 - \left(\sum_{i=1}^{n} y_i\right)^2}}$$

$$\sum_{i=1}^{n} x_i = 70 \quad \sum_{i=1}^{n} x_i^2 = 728 \quad \sum_{i=1}^{n} x_iy_i = 676$$

$$\sum_{i=1}^{n} y_i = 63 \quad \sum_{i=1}^{n} y_i^2 = 651 \quad n = 7$$

$$r = \frac{7\times 676 - 70\times 63}{\sqrt{\left[7\times 728 - 70^2\right]} \times \sqrt{\left[7\times 651 - 63^2\right]}}$$

$$= \frac{4732 - 4410}{\sqrt{[5096 - 4900]} \times \sqrt{[7\times 651 - 3969]}}$$

$$= \frac{322}{\sqrt{196} \times \sqrt{588}} = \frac{322}{14\times 24.25} = \frac{322}{339.5} = 0.95$$

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